B.Tech.(Computer Engg.) Vth Semester Examinations

Session 2021-22

Time: 1 hour Object Oriented Programming (CEN - 504) Sessional-II Max. Marks: 15

Note: Attempt All Questions

S. No.	Questions	Marks	Course Outcome
1.	Design a *Complex* class representing complex numbers. A complex number in mathematics is defined as $x + iy$ where x defines the real part of the number and y is the imaginary part. The letter i represents the square root of -1 (which means $i2$ is -1). Include operator functions to overload the operators +=, -=, *=, /= and the << operator for the class. Here << operator should be used for printing the results of complex number operation. $(x_1 + i\,y_1) + (x_2 + i\,y_2) = (x_1 + x_2) + i\,(y_1 + y_2)$ $(x_1 + i\,y_1) - (x_2 + i\,y_2) = (x_1 - x_2) + i\,(y_1 - y_1)$ $(x_1 + i\,y_1) * (x_2 + i\,y_2) = (x_1x_2 - y_1y_2) + i\,(x_1y_2 + x_2y_1)$ $(x_1 + i\,y_1) / (x_2 + i\,y_2) = ((x_1x_2 + y_1y_2) + i\,(-x_1y_2 + x_2y_1)) /$ denominator where denominator $= x_2^2 + y_2^2$	4	CO2
2.	Consider an example of a bookshop which sells books and video tapes. These two classes are inherited from the base class called media. The media class has command data members such as title and publication. The book class has data members for storing a number of pages in a book, and the tape class has the playing time in a tape. Each class will have member functions such as read() and show(). In the base class, these members have to be defined as virtual functions. Write a program which models the class hierarchy for the bookshop and processes objects of these classes using pointers to the base class.	4	CO3
3.	Explain following oops concepts with suitable example: (a) Pure Virtual Function (b) Pointers to Derived Class Object (c) Virtual Destructor (d) Overloading through friend function	4	CO3
4.	Write a program to overload [] operator for the following scenario: Create a class AccountBook that contains account holder details such as name and account number. Take input for 5 account holders in the account table. When we enter account number, then the program prints account holder name while entering of account holder name, it prints account number of holder.	3	CO2